CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 19

16 BIT SUBTRACTION USING USING 8085

AIM: To write and execute an 8085 assembly language program to find 16 bit addition and store result in memory.

ALGORITHM

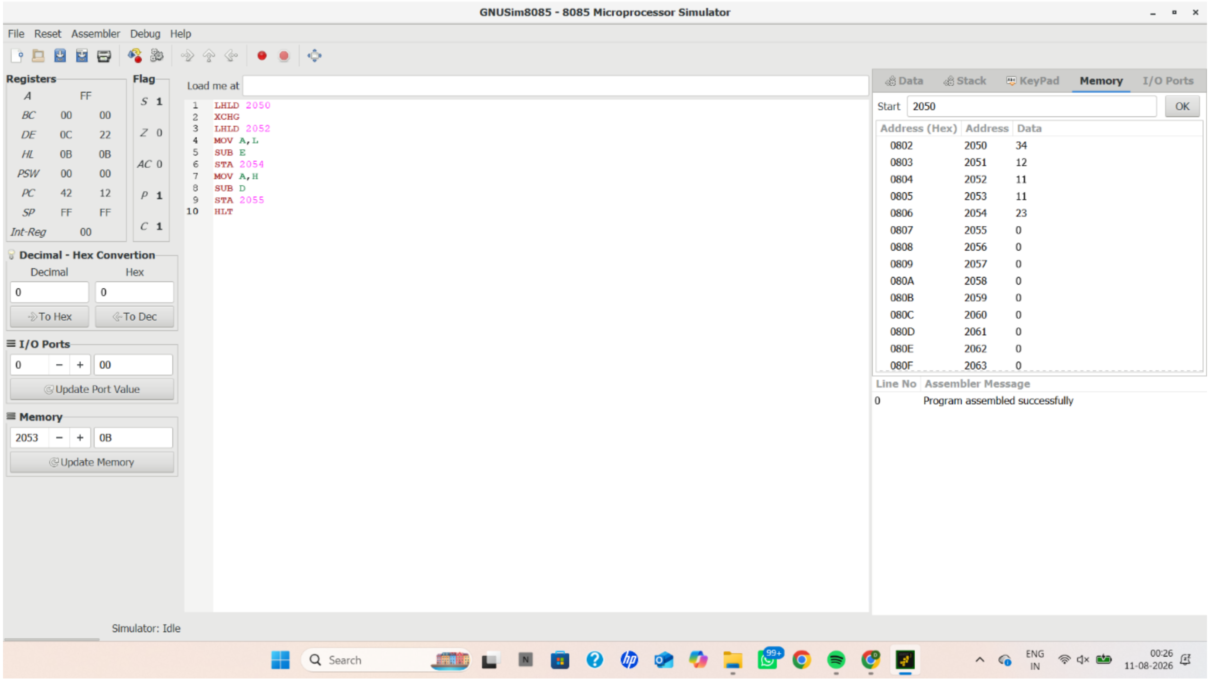
1. Load the lower byte of the first number.
2. Subtract the lower byte of the second number using SUB.
3. Store the result.
4. Load the higher byte of the first number.
5. Subtract the higher byte of the second number along with borrow using SBB.
6. Store the higher-byte result.
7. Stop the program.

CODE:

```
LDA 2050H
MOV B,A
LDA 2052H
MOV C,A
MOV A,B
SUB C
STA 2054H
LDA 2051H
MOV B,A
LDA 2053H
MOV C,A
MOV A,B
SBB C
STA 2055H
```

HLT

OUTPUT:


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
Flag
Load me at
LHLD 2050
XCHG
LHLD 2052
MOV A,L
SUB E
STA 2054
MOV A,H
SUB D
STA 2055
HLT
Data Stack KeyPad Memory I/O Ports
Start 2050 OK
Address (Hex) Address Data
0802 2050 34
0803 2051 12
0804 2052 11
0805 2053 11
0806 2054 23
0807 2055 0
0808 2056 0
0809 2057 0
080A 2058 0
080B 2059 0
080C 2060 0
080D 2061 0
080E 2062 0
080F 2063 0
Line No Assembler Message
0 Program assembled successfully
Decimal - Hex Convertion
Decimal Hex
To Hex To Dec
I/O Ports
Update Port Value
Memory
2053 0B
Update Memory
Simulator: Idle
Search
00:26
11-08-2026